B.E. Even Semester Midterm Examination, 2024-25

Subject: Discrete Mathematics | **Paper Code:** PCC-IT401 | **Full Marks:** 15 | **Time:** 1 Hr.

Q1. Define a bipartite graph. Draw $K_{3,3}$.

Answer: A **bipartite graph** is a graph whose vertices can be divided into two disjoint and independent sets U and V such that every edge connects a vertex in U to one in V. There are no edges connecting vertices within the same set. $K_{3,3}$ has two sets of vertices, $U = {u_1, u_2, u_3}$ and $V = {v_1, v_2, v_3}$. Every vertex in U is connected to every vertex in V.

Q2. Show that $(\mathbb{R},+)$ forms a group but $(\mathbb{R},\cdot)$ does not form a group.

Answer: 1. $(\mathbb{R},+)$ is a group: Satisfies Closure, Associativity, Identity (0), and Inverse (-a). **2. $(\mathbb{R},\cdot)$ is NOT a group:** It fails the **Inverse property**. For the element $0 \in \mathbb{R}$, the inverse $1/0$ is undefined.

Q3. Define a group with an example. Give an example of a monoid which is not a group with reason.

Answer: Definition: A group $(G, *)$ is a set G combined with a binary operation $*$ satisfying Closure, Associativity, Identity, and Inverse. Example: $(\mathbb{Z}, +)$. **Monoid not a group:** $(\mathbb{N}, +)$ where $\mathbb{N} = {0, 1, 2, ...}$ is a monoid. Identity is 0. It's not a group because elements (like 5) lack an additive inverse (-5) in $\mathbb{N}$.

Q4. Draw a complete graph with five vertices. Verify Handshaking theorem for the graph.

Answer: A complete graph K_5 has 5 vertices connected to every other vertex once. * Vertices ($|V|$) = 5. Edges ($|E|$) = $5(4)/2 = 10$. Degree of each = 4. * **Handshaking Theorem:** Sum of degrees = $2|E|$. LHS: $5 \times 4 = 20$. RHS: $2 \times 10 = 20$. Verified.

Q5. Draw a graph G with six vertices. Find four subgraphs of this graph G with at least four vertices.

Answer: Let graph G be a cycle graph C_6 defined by vertices $V = {1, 2, 3, 4, 5, 6}$. Four subgraphs: 1. Path of 5 vertices: Remove vertex 6. 2. Path of 4 vertices: Remove vertices 5 and 6. 3. Disconnected 4 vertices: Remove vertices 1 and 4. 4. G without one edge (Spanning subgraph with 6 vertices).